

Here are **3 practice tasks** (one for each constructor type) using different real-time scenarios. These are suitable for students learning constructors.

Task 1: Default Constructor

Problem: Book Information

Create a class named `Book` with the following instance variables:

- `bookId`
- `title`
- `author`
- `price`

Requirements

- Create a **default constructor**.
- Inside the constructor, initialize the following values:
 - Book ID: `101`
 - Title: `"Java Programming"`
 - Author: `"James Gosling"`
 - Price: `650`
- Create a `display()` method to print all book details.
- Create one object and display the information.

Expected Output

```
None
Book ID : 101
Title   : Java Programming
Author  : James Gosling
Price   : 650
```

Task 2: Parameterized Constructor

Problem: Employee Payroll System

Create a class named `Employee` with the following instance variables:

- `empId`
- `empName`
- `department`
- `salary`

Requirements

- Create a **parameterized constructor** to initialize all values.
- Create a `display()` method.
- Create **three employee objects** with different data.
- Display all employee details.

Sample Data

Employee ID	Name	Department	Salary
101	Ravi	HR	35000
102	Priya	Developer	55000
103	Kiran	Testing	42000

Expected Output

None

Employee ID : 101

Name : Ravi

Department : HR

Salary : 35000

Employee ID : 102

Name : Priya

Department : Developer

Salary : 55000

Employee ID : 103

Name : Kiran

Department : Testing

Salary : 42000

Task 3: Copy Constructor

Problem: Laptop Inventory

Create a class named `Laptop` with the following instance variables:

- `brand`
- `model`
- `price`

Requirements

- Create a **parameterized constructor** to initialize the laptop details.
- Create a **copy constructor** that copies the values from another `Laptop` object.
- Create a `display()` method.
- Create one laptop object using the parameterized constructor.
- Create another laptop object using the **copy constructor**.
- Display both objects.

Sample Data

None

```
Brand : Dell  
Model : Inspiron 15  
Price : 62000
```

Expected Output

None

Laptop 1

```
Brand : Dell  
Model : Inspiron 15  
Price : 62000
```

Laptop 2

```
Brand : Dell  
Model : Inspiron 15  
Price : 62000
```

4.Scenario-Based Practice Question

A newly opened **ABC Electronics Showroom** wants to develop a Java application to manage its mobile inventory. Whenever a new mobile is added to the showroom, the system should be able to create mobile objects in different ways based on the available information.

If no details are provided, the system should automatically create a mobile object with the showroom's default mobile information. If the showroom manager enters the complete details of a mobile, the system should create the object using those values. Sometimes, the showroom receives multiple mobiles with exactly the same specifications. Instead of entering the same information repeatedly, the system should create a new mobile object by copying the details of an existing mobile.

Design a Java program by creating a class named **Mobile** with appropriate instance variables. Implement a **default constructor**, a **parameterized constructor**, and a **copy constructor**. Also, create a **display()** method to print the mobile details.

In the **main()** method:

- Create the first mobile object using the **default constructor**.
- Create the second mobile object using the **parameterized constructor** with your own values.
- Create the third mobile object by copying the second object using the **copy constructor**.
- Display the details of all three mobile objects.
- Finally, change the price of the third mobile object and display the details of both the second and third objects to verify whether changing one object affects the other.

===== MOBILE DETAILS =====

Mobile 1 (Default Constructor)

Mobile ID : 1001
Brand : Samsung
Model : Galaxy A35
Price : 28000.0

Mobile 2 (Parameterized Constructor)

Mobile ID : 1002
Brand : OnePlus
Model : Nord CE 5
Price : 24999.0

Mobile 3 (Copy Constructor)

Mobile ID : 1002

Brand : OnePlus
Model : Nord CE 5
Price : 24999.0

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After Updating the Price of Mobile 3

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Mobile 2 Details

Mobile ID : 1002
Brand : OnePlus
Model : Nord CE 5
Price : 24999.0

Mobile 3 Details

Mobile ID : 1002
Brand : OnePlus
Model : Nord CE 5
Price : 23000.0